Mwangi ndungu

79

BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A

TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

WRITTEN ASSESSMENT

CAT 1

TIME: 2 HOURS

INSTRUCTIONS TO THE CANDIDATE

1. *Answer all questions.*
2. *Write your answers clearly and legibly.*
3. *Show all working where calculations are required.*
4. *Where diagrams are required, use appropriate electrical/electronic symbols.*
5. *The marks allocated to each question are shown in brackets.*

1.

a) Define an electrical circuit. (2 marks)

b) State four components of a basic electrical circuit. (4 marks)

c) State the SI unit for each of the following electrical quantities: (5 marks)

Electrical quantity	SI unit
EMF	
Current	
Power	
Resistance	
Energy	

2.

a) What is meant by electric current? (2 marks)

b) A resistor has a resistance of 10 Ω and carries a current of 2 A. Calculate the voltage ac resistor. (3 marks)

c) Distinguish between AC and DC. (2 marks)

d) Give one application of DC. (1 mark)

3.

a) Determine the resistance value of a resistor with the following colour bands:

Brown – Black – Red – Gold. (3 marks)

b) Determine the resistance value and tolerance of a resistor with the following colour bands:

Red – Violet – Orange – Gold. (3 marks)

c) State the function of the fourth colour band on a resistor. (2 marks)

4.

a) Define a semiconductor. (2 marks)

b) Name two semiconductor materials. (2 marks)

c) With an aid of a diagram differentiate between P-type and N-type semiconductor. (4 marks)

d) State the two biasing conditions of a PN-junction diode. (2 marks)

5.

a) With the aid of a labelled V–I characteristic curve, describe the behaviour of a PN-junction diode in forward bias and reverse bias. (6 marks)

b) State two applications of a PN-junction diode. (2 marks)

6.

a) What is a transistor? (2 marks)

b) Name the two types of bipolar junction transistors. (2 marks)

c) State the three terminals of a bipolar junction transistor. (3 marks)

d) Give three applications of transistors in electronic circuits. (3 marks)

7.

a) Define an integrated circuit (IC). (2 marks)

b) State three advantages of using integrated circuits in electronic equipment. (3 marks)

c) State one characteristic of each of the following components: (4 marks)

i. Resistor

ii. Capacitor

d) Classify computer memory according to the technology used by giving examples for each: (2 marks)

i. Semiconductor memory — _____

ii. Magnetic memory — _____

iii. Optical memory — _____

(1a) An Electrical circuit is a complete path through which Electric current flows from power source, through the components and back to the source. (2)

(b) (i) power source
(ii) conducting wires
(iii) Load
(iv) Switch (4)

(c) EMF → volt (V)
(ii) current → Ampere (A)
(iii) Power → watt (W)
(iv) Resistance → ohm (Ω)
(v) Energy → Joule (J) (5)

(2a) Electric current is the flow of ~~electron~~ electric charge through a conductor. It is measured in amperes (A) (2)

(b)
$V = IXR$
$I = 2A$
$R = 10\Omega$
$V = 2 \times 10$ (3)
$V = 20V$

(c) AC → current changes its direction periodically while DC → current flows in one direction only. (2)

(d) DC is used to power electronic devices such as mobile phones and computers (1)

(3a)
> Brown → 1
> Black → 0
> Red → X 100 (1)
> Gold → ± 5%

7

20

(3b) $27 \times 1000 \pm 5\%$

$27{,}000 \times \frac{5}{100}$

$= 1{,}350$

max = $13{,}50 + 27000$

$= 28{,}350$

Min = $1350 - 27000$

$= 25650$ (3)

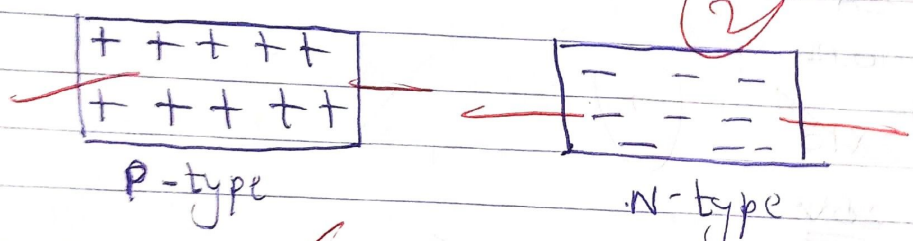
(3c) The 4th color band shows the tolerance of the resistor. It tells us how much the actual resistance can vary from its state value (2)

(4a) ~~A~~ is a material whose ability to conduct electricity is between that of a conductor and an insulator (2)

(4b) Silicon

> Germanium

(4c) p-type is made by adding a ~~tri~~valent impurity while N-type mad by adding pentavalent impurity.

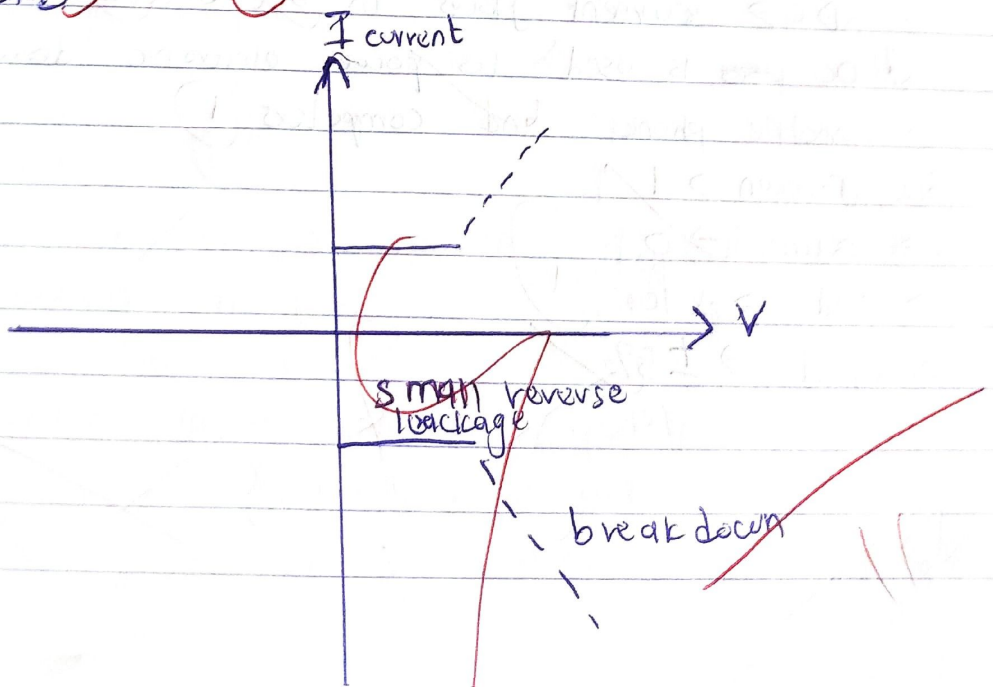


(2)

(4d) Forward bias

> Reverse bias (2)

(5a)

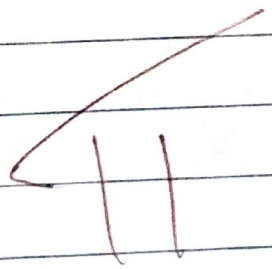


-80 dBm is far below -30 dBm. Can cause equipment reboot
switch 5 has an open circuit - cut and replace the connector
-82 dBm is a very weak signal - relocate an access point near the meeting room

(a) Switch(config)# vlan 10
Switch(config-vlan)# name FINANCE
Switch(config-vlan)# exit
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10

(b) The VLANs are separate broadcast domains and a layer 2 switch won't forward traffic between them. Use inter-VLAN routing.

- Layer 3 switch - enable ip routing and create SVIs
Router-on-a-stick - trunk link a router with subinterfaces



(5b) Rectification - converting AC to DC
> Signal detection in electronic circuits
(6a It is a semiconductor device used to amplify or switch Electronic signals
(6h NPN transistor
> PNP transistor
(6d Amplification of signals
> Electronic switching
> oscillators circuits
6e) Emitter
> Base
> collector
(7a Is a small electronic device containing many components such as transistors, resistors and capacitors on a single semiconductor chip
(7b small in size
> consume less power
> Reliable and fewer connections, making them less likely to fail.
(7c Resistor → It opposes or limits the flow of electric current
> capacitor → Stores electrical charge temporarily.
> Diode → Allows current to flow mainly in one direction.
> Inductor → Stores electrical energy in the form of a magnetic field.
(7d An intergrated circuit is used in computers and ot electronic devices to perform processing and contr functions.
(8 a Is a part of a computer that stores data, instru and information for processing or future use.
(8 b Primary memory
secondary memory

27

(8c Ram
> ROM
(8d Hard disk
> Flash disk
(8e primary memory is faster while secondary is slower
> primary memory is directly accessed by the cpu, while secondary storage is mainly used for long term storage
(9a Registers
> cache memory
> Main memory
> Secondary storage
> Tertiary storage
(9b Registers → They are very fast located inside the cpu
> cache memory → is a small high speed memory used to store frequent needed data
> main memory → it stores programs and data currently being used by the computer
> secondary storage → it provides long term storage of data & files
> Tertiary storage → Its mainly used for large-scale storage and backup for data.

(10a

Ram	Rom
> Its volatile memory	> Is non-volatile memory
> It is used to store data & programs currently being used	> It keeps its contents even when there is no p
> can be read from and written to easily	> is mainly read-only or has limited writ

(10b Its volatile
> It allows both reading and writing of data.
(10c Its non-volatile
> It retains data even when power is switched off
10 D) Semiconductor memory → RAM / ROM
> Magnetic memory → Harddisk / magnetic tape
> optical memory → CD / DVD